Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

2081

Bachelor of Science in Computer Science and Information Technology

Course Title: **Mathematics II**

Code No: **MTH168** Semester: **2nd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Define augmented matrix with an example. Find the general solution of the linear system whose augmented matrix is $\begin{bmatrix} 1 & 6 & 2 & -5 \\ -1 & 0 & 3 & 1 \\ 0 & -1 & -2 & 3 \end{bmatrix}.$

2. (a) Let $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, and define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(x) = Ax$. Find the image under T of $u = \begin{bmatrix} 1 \\ -3 \end{bmatrix}, v = \begin{bmatrix} a \\ b \end{bmatrix}$.(b) Prove that a map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (y, x)$ is linear.

3. Define inverse of a matrix. Find the inverse of a matrix $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ -3 & 4 & 8 \end{bmatrix}$, if it exists.

Section B

Attempt any Eight questions[8×5=40]

4. Verify that $(AB)^{-1} = B^{-1}A^{-1}$ if $A = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$.

5. Find LU factorization of $\begin{bmatrix} 2 & 5 \\ 6 & -7 \end{bmatrix}$.

6. Compute the determinants by cofactor expansions $\begin{vmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{vmatrix}$.

7. Show that the column vectors $u = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, v = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, w = \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$ and $x = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$ is x in span $\{u, v, w\}$.

8. Let $B = \{b_1, b_2\}$ and $C = \{c_1, c_2\}$ be bases for a vector space V, and suppose $b_1 = 6c_1 - 2c_2$ and $b_2 = 9c_1 - 4c_2$, then(a) find the change of coordinates matrix from B to C.(b) find $[x]_c$ for $x = -3b_1 + 2b_2$.

9. Let $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$. Find the eigenvalues and eigenvectors of A.

10. Determine the least squares error in the least-square solution of $Ax = b$, where $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$.

11. Prove that the binary operation * defined on Z by letting $m * n = m + n + 1$ iscommutative and associative.

12. Show that $(Q, +, .)$ forms a ring.